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THESIS RESEARCH PLAN

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ABBREVIATIONS

ENSFM	Efficient Non-Sampling Factorization Machines
RS	Recommender systems
CF	Collaborative filtering
CB	Content-based filtering
FM	Factorization machines
CTR	Click-through rate
BPR	Bayesian Personalized Ranking

USE OF ARTIFICIAL INTELLIGENCE IN THIS THESIS RESEARCH PLAN

In this research, I have used ChatGPT as a tool for brainstorming, language refinement, and grammar correction. The ideas and content presented in this thesis research plan are entirely my own, and ChatGPT has only been used to improve clarity and ensure grammatical accuracy. Every section included in this research plan has been carefully reviewed, fact-checked, and verified by me before being finalized.

I have ensured that the use of artificial intelligence does not compromise the originality of my work. While AI-assisted tools have helped refine the wording, all concepts, arguments, and methodologies originate from my own research and understanding. I have used AI responsibly, ensuring that my research adheres to ethical standards and maintains full respect for academic integrity and copyright. I have not used artificial intelligence in writing this section.

1 INTRODUCTION

Recommender systems have become a critical component in modern e-commerce, enabling personalized experiences and improving customer engagement. Internationally, advancements in deep learning and factorization-based models have significantly enhanced recommendation accuracy. However, traditional approaches often rely on sampling-based optimization, which can lead to inefficiencies and suboptimal performance, particularly on large-scale datasets. The Efficient Non-Sampling Factorization Machines (ENSFM) approach, introduced in 2020, addresses this issue by eliminating the need for negative sampling, leading to more stable and effective ranking performance. While ENSFM has shown promise in previous studies, its application to large, real-world e-commerce datasets remains largely unexplored.

This thesis aims to evaluate ENSFM's scalability, efficiency, and accuracy when applied to modern, large-scale e-commerce datasets from Yelp2018, Amazon Books Reviews 2023 dataset. The findings will contribute to the development of recommendation systems, particularly in commercial applications in Finland, where data-driven personalization is crucial for competitive advantage. Through experimentation and benchmarking, this study seeks to enhance the adaptability of ENSFM for large-scale, high-performance recommendation systems, bridging the gap between academic research and real-world deployment.

2 PURPOSE, OBJECTIVES AND RESEARCH QUESTIONS/ TASKS

2.1 Purpose:

The primary purpose of this thesis is to investigate the feasibility of deploying the Efficient Non-Sampling Factorization Machines (ENSFM) model within the context of modern, large-scale e-commerce applications. While the original ENSFM approach effectively addressed inefficiencies of sampling-based learning, its performance characteristics and practical applicability when using datasets significantly larger and more complex than its initial benchmarks remain largely unexplored. This research seeks to bridge this gap by assessing how well the existing ENSFM model adapts to and performs under the demands of current real-world data environments, ultimately aiming to provide a deeper, practical understanding of its suitability for contemporary commercial recommender systems. The expected outcome is a more practical, efficient, and user-friendly recommendation system that can be seamlessly integrated into large-scale commercial applications.

2.2 Objective:

To achieve this purpose, the specific objectives involve conducting a comparative performance analysis by applying the ENSFM model to new, large-scale e-commerce datasets and evaluating its results against the performance achieved on the original benchmark datasets. Furthermore, controlled experiments will be designed and executed to identify key factors within these new, larger datasets that significantly influence the model's effectiveness. This process includes rigorously testing the model's applicability and behaviour in these scaled-up scenarios, with a clear focus on assessing feasibility rather than achieving optimal performance metrics.

2.3 Research Questions / Tasks:

To achieve these goals, this study seeks to answer the following key research questions:

- Question 1: Does the ENSFM model work well on the new dataset?

- Question 2: Which factors in the new datasets strongly influence the model's performance?

By addressing these research questions, this thesis will evaluate the performance and practical viability of the ENSFM model when applied to modern, large-scale datasets. Critically, it will also identify key dataset characteristics that significantly influence the model's effectiveness. The findings will therefore provide valuable insights into the challenges and considerations for deploying ENSFM, contributing a deeper, evidence-based understanding necessary for assessing its suitability in real-world, data-intensive e-commerce environments.

3 DATABASE

Recommender systems (RS) play a crucial role in modern e-commerce by providing personalized product suggestions to users based on their past behavior, preferences, and contextual information. Traditionally, collaborative filtering (CF), content-based filtering (CB), and hybrid models have been widely used to improve recommendation accuracy. However, as datasets grow exponentially in size and complexity, newer approaches leveraging factorization machines (FM) and deep learning models have emerged as more scalable and effective solutions. This thesis focuses on Efficient Non-Sampling Factorization Machines (ENSFM), an advanced RS approach that eliminates the need for negative sampling, leading to more accurate and computationally efficient recommendations.

3.1 Key Concepts and Their Connection to This Thesis:

3.1.1 Factorization Machines (FM)

Factorization Machines (FM) are a powerful machine learning model designed to handle high-dimensional and sparse data, making them particularly well-suited for recommender systems. Unlike traditional collaborative filtering (CF), which primarily focuses on user-item interactions, FM extends its capability to model interactions between all available features, including contextual information such as time, location, and user demographics. This flexibility allows FM to be used not only for recommendations but also in areas like predictive analytics and click-through rate (CTR) prediction.

One of FM's key advantages is its ability to efficiently learn second-order feature interactions without explicitly storing all possible combinations, which would be computationally expensive. This makes it more scalable compared to traditional matrix factorization methods. FM can work with sparse datasets, where most user-item interactions are missing, making

it particularly useful for real-world applications where data sparsity is a major challenge. However, a common limitation of FM in large-scale recommendation tasks is its reliance on negative sampling, a technique used to reduce the computational cost of training. While negative sampling speeds up training, it can introduce bias and lead to suboptimal ranking performance in recommendation tasks.

3.1.2 Efficient Non-Sampling Factorization Machines (ENSFM)

Efficient Non-Sampling Factorization Machines (ENSFM) are an advanced variation of FM designed to eliminate the need for negative sampling, thereby improving ranking accuracy and computational efficiency. Unlike traditional FM, which relies on sampling strategies to approximate missing interactions, ENSFM processes all interactions directly, allowing for more precise optimization. This makes ENSFM more stable, scalable, and accurate, particularly in context-aware recommendation tasks where feature interactions play a crucial role.

ENSFM has been proposed as a solution to the ranking performance issues caused by negative sampling in FM-based models. By avoiding the bias introduced by sampling, ENSFM achieves more reliable predictions, making it suitable for applications that require highly accurate ranking and real-time recommendation processing. While early research has demonstrated that ENSFM outperforms traditional FM in controlled experimental settings, its applicability to large-scale, modern e-commerce datasets remains an open question. This thesis aims to evaluate whether ENSFM can maintain its efficiency, speed, and recommendation quality when applied to real-world commercial applications with millions of user-item interactions.

Additionally, this research refines ENSFM's training output format, ensuring that its predictions and model diagnostics are more interpretable and accessible for practitioners. By making ENSFM's results easier to analyze, this study enhances its usability in real-world machine learning

pipelines, allowing developers to integrate, fine-tune, and optimize ENSFM more effectively.

3.2 Existing Research on the Topic

The foundation of this study is built upon basic recommender system principles and the findings of the ENSFM paper. Recommender systems have traditionally relied on collaborative filtering (CF) and content-based filtering (CB) to generate personalized recommendations.

- **Collaborative Filtering (CF):** CF is one of the most widely used approaches, relying on user-item interactions to generate recommendations. It can be categorized into memory-based CF, which computes similarity between users or items, and model-based CF, which utilizes machine learning techniques such as matrix factorization to learn latent user and item representations. A key limitation of CF is the cold start problem, where new users or items lack sufficient interaction history, making accurate recommendations difficult.
- **Content-Based Filtering (CB):** Unlike CF, CB recommends items based on their attributes rather than user interactions. This approach analyzes item features and user preferences to generate suggestions. While CB is effective in cold start scenarios, it struggles with feature sparsity and requires rich metadata to perform well.
- **Hybrid Approaches:** To overcome the limitations of both CF and CB, hybrid recommendation models combine multiple techniques, leveraging the strengths of each approach.

Factorization-based models such as FM have been widely adopted to enhance CF-based recommender systems by modeling feature interactions more effectively. However, sampling-based FM models, including those using Bayesian Personalized Ranking (BPR-FM), have limitations

in ranking accuracy due to their reliance on incomplete negative interactions. The ENSFM approach, as proposed in the ENSFM paper, directly addresses this limitation by removing negative sampling while maintaining computational efficiency.

Prior research on ENSFM has demonstrated that removing negative sampling leads to more stable and effective rankings in recommendation tasks. However, the original study focused on specific datasets, and its applicability to larger, more complex e-commerce datasets has not been explored. This thesis extends the ENSFM research by applying it to modern e-commerce datasets and analyzing whether it retains its efficiency, speed, and ranking performance in a larger, more dynamic environment.

Additionally, previous work on ENSFM has not addressed the interpretability of its training output. Many machine learning practitioners and researchers rely on clear, structured outputs to fine-tune models and diagnose performance issues. By improving ENSFM's training output format, this research ensures that its results are more readable, actionable, and accessible for those implementing it in real-world applications.

Through this research, ENSFM's effectiveness will be critically evaluated on new datasets, focusing on whether it continues to perform as efficiently and accurately as reported in prior work. If deviations are observed, this study will investigate the underlying factors contributing to performance differences, providing insights into how ENSFM can be further optimized for large-scale recommender systems.

4 RESEARCH METHODS AND MATERIALS

This chapter presents the research methodology used to evaluate ENSFM in large-scale recommender systems. The study follows an experimental and data-driven approach, applying ENSFM to two large-scale e-commerce datasets to assess its performance in terms of accuracy, ranking quality, and computational efficiency. The methods for data collection, analysis, and model development are chosen based on established practices in machine learning and recommender system evaluation.

4.1 Data Collection Method

To ensure that the datasets used in this study are suitable for evaluating Efficient Non-Sampling Factorization Machines (ENSFM), a thorough selection process was conducted. The original ENSFM paper utilized datasets with structured user-item interactions, explicit and implicit feedback, and metadata features, making it essential to find datasets with similar characteristics. Additionally, the datasets needed to be publicly accessible and sourced from reliable platforms, ensuring that the research could be reproduced and validated by others.

After reviewing multiple datasets, two large-scale e-commerce datasets were chosen:

4.2 Yelp2018 dataset

- Contains nearly 7 million interactions (reviews/ratings) from almost 2 million users evaluating over 150,000 businesses (items).
- Provides detailed user-business interaction data, likely including ratings, timestamps, and potentially user/business metadata derived from its 9 distinct fields, making it suitable for modeling user preferences and rating prediction.

- The dataset structure, encompassing over 5.9 million feature instances, allows for analysis based on user profiles, business characteristics, and the context of the interaction, making it ideal for evaluating recommendation systems and understanding factors driving user engagement with local businesses.

4.3 Amazon Books Reviews 2023 dataset

- Features an extremely large scale, containing over 145 million reviews generated by 10.3 million users for 4.4 million distinct books listed on Amazon, covering interactions from May 1996 to September 2023.
- Provides extensive user-book interaction data (reviews, ratings) alongside rich contextual information, including separate detailed metadata for books (e.g., title, author, categories), making it highly suitable for modeling reading preferences and purchase behavior in a major e-commerce setting.
- The dataset's structure, spanning 10 fields across two large JSONL files and encompassing over 29.5 million feature instances, allows for in-depth feature engineering that combines user behavior with specific book attributes. This makes it an ideal testbed for evaluating recommendation systems under conditions of large scale and rich item context, characteristic of real-world e-commerce platforms.

The Yelp2018 and Amazon Books Reviews 2023 datasets were selected due to their large scale, the presence of rich user and item contexts, massive interaction volumes, and their representation of real-world scenarios suitable for testing the feasibility of the ENSFM model.

As the original ENSFM preprocessing steps were undocumented, a custom, structured preprocessing pipeline was developed and applied to ensure dataset compatibility with the model. Key steps include:

- Filtering: Setting thresholds (via arguments) for minimum interactions per user and item, then filtering users, items, and interactions accordingly. Outlier handling based on item features may also be applied.
- ID Assignment & Feature Engineering: Assigning unique sequential integer IDs to valid users and items. Extracting relevant features (e.g., average ratings, price, category IDs) from metadata and interactions, followed by numerical scaling/transformation to ensure distinct value ranges.
- Sequence Handling & Sampling: Sorting each user's interactions chronologically. Optionally applying category balancing (via upsampling) or stratified user subsampling based on activity levels.
- Re-indexing & Formatting: Performing global re-indexing to create dense, sequential IDs for users and items remaining after all steps (especially for 'unified' mode processing). Formatting user and item features into the required input string structure (e.g., "ID-feature1-feature2...").
- Data Splitting: Dividing each user's chronologically ordered interactions into training and testing sets, typically designating the final interaction for the test set.

This multi-stage preprocessing pipeline, with code available on GitHub, yields clean, structured data formatted specifically for the ENSFM model, enabling a reliable evaluation of its performance on these large-scale datasets.

4.4 Data Analysis Method

To evaluate ENSFM’s performance, quantitative analysis techniques are applied. The key evaluation metrics include:

- **Hit Ratio (HR@K):** Measures the percentage of cases where the correct item appears in the top-K recommendations.
- **Normalized Discounted Cumulative Gain (NDCG@K):** Evaluates ranking performance by considering the position of relevant items in the recommendation list.
- **Training Time and Inference Speed:** Assesses ENSFM's computational efficiency when applied to large-scale datasets.

Unlike prior work that benchmarks ENSFM against multiple models, this research directly compares ENSFM's performance on new datasets with its original reported results. The goal is to determine:

1. Does ENSFM maintain its effectiveness on larger, real-world datasets?
2. Which factors on the new datasets affect to the performance of the model ENSFM?

Visualization techniques (such as training curves, ranking distributions, and error heatmaps) are used to interpret how ENSFM adapts to different dataset characteristics

4.5 Development Method

The research employed a systematic methodology to evaluate the feasibility of the ENSFM model on new, large-scale datasets. This involved updating the technical environment, validating the original model's implementation, carefully selecting and preparing new datasets using a custom pipeline, and conducting standardized training and evaluation.

4.5.1 System Environment and Code Validation

The initial phase involved updating the original ENSFM codebase environment from Python 2.7 to 3.9 and upgrading the TensorFlow library to a newer version. To ensure the integrity of the original model logic

after updates, the original experiments were re-executed using the authors' datasets on multiple platforms (local PC, Kaggle, Colab). Consistency in performance metrics (HR, NDCG) with the original paper's results confirmed the successful validation of the updated codebase.

4.5.2 Dataset Selection

New datasets, Yelp2018 and Amazon Books Reviews 2023, were identified by systematically reviewing literature that cited the original ENSFM study.

Selected datasets were thoroughly examined (documentation, structure) to confirm their suitability based on scale, context availability, interaction volume, and compatibility with ENSFM requirements for testing feasibility in real-world scenarios.

4.5.3 Data Preprocessing

As original preprocessing steps were undocumented, a custom, configurable data preprocessing pipeline was developed (code available on GitHub) specifically for this thesis.

This pipeline includes:

- Configurable filtering of users, items, and interactions based on minimum activity thresholds and potentially outlier removal.
- Assigning unique sequential integer IDs to valid users and items.
- Extracting relevant user/item features from metadata and interactions, followed by numerical scaling and transformation into distinct value ranges.
- Sorting user interactions chronologically for time-based splitting.
- Optional steps for category balancing (via up-sampling) or stratified user subsampling.
- Global re-indexing to ensure dense IDs (especially for multi-category 'unified' mode).

- Formatting features into the specific dash-separated string structure required by the ENSFM model input (e.g., "ID-feature1-feature2...").

4.5.4 Model Training and Evaluation

- The validated ENSFM model code, including the LoadData and ENSFM classes provided by the original authors, was utilized for training and evaluation.
- The preprocessed Yelp2018 and Amazon Books Reviews 2023 datasets (formatted train.csv and test.csv) were loaded.
- The model was trained using a predefined set of hyperparameters (batch size: 512, embedding size: 64, learning rate: 0.05, epochs: 501, etc.).
- Performance was evaluated periodically during training using standard Top-K recommendation metrics: Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG) at K values of 5, 10, and 20.

This structured approach ensured a reliable assessment of ENSFM's applicability and performance on modern, large-scale datasets. By validating the baseline, carefully selecting and preparing new data via a defined pipeline, and using standardized evaluation procedures, the research provides practical insights into the model's behavior in contemporary scenarios.

5 ETHICS AND RELIABILITY

5.1 Research Ethics and Data Protection

This research does not involve direct human participants or the processing of sensitive personally identifiable information (PII). The datasets used—Yelp2018 and Amazon Books Reviews 2023—were obtained from sources making them available for academic research purposes. This availability ensures they are legally and ethically suitable for this study. Consequently, given the nature of the data and its source, no special research permits were required. The study strictly adheres to the terms of use specified by the respective dataset providers and original research studies, ensuring responsible and ethical data handling throughout the project.

Regarding personal data processing and protection, the datasets utilized do not contain sensitive personal details such as names, addresses, or payment information. The analysis focuses solely on user interaction data (reviews, ratings), associated metadata, and pseudonymized user/item identifiers, significantly minimizing any risk of re-identification. Although confidentiality risks are inherently low with such data, all data processing and storage procedures adhere to data protection best practices. The datasets are stored securely, and access is restricted solely for the research purposes outlined in this thesis, preventing misuse.

5.2 Reliability and Validity of the Research

To ensure the reliability and validity of this research, I strictly follow the Scrum methodology, setting clear deadlines, structured work schedules, and iterative development cycles for each phase of the thesis. By maintaining a well-organized workflow, the likelihood of deviating from the

objectives outlined in the research plan is minimized. Each stage of the project is carefully planned, reviewed, and adjusted based on progress, ensuring that all research goals are met effectively.

Additionally, I systematically document all research activities, experiment results, and insights in Notion, where I also store references, notes, and previous findings. This structured approach makes it easy to track past work, retrieve relevant materials, and maintain research consistency. With this level of organization, the research process remains highly transparent, reproducible, and efficient, reinforcing the reliability of the final results.

6 SCHEDULE AND RESOURCE PLAN

6.1 Schedule:

Given that the dataset selection and ENSFM implementation have already been completed, the remaining tasks are structured into a two-month timeline to ensure efficient completion of the thesis:

- **Week 1 – Week 2**: Data preprocessing, refining training output format, and setting up evaluation metrics.

- **Week 3 – Week 4**: Running experiments, analyzing ENSFM performance on new datasets, and optimizing model parameters.

- **Week 5 – Week 6**: Comparing results with the original ENSFM paper, interpreting findings, and documenting key insights.

- **Week 7 – Week 8**: Finalizing thesis writing, making revisions, and preparing for submission.

6.2 Resource Plan:

No additional computational resources or external funding are required for this research. However, any guidance or insights from my supervisor, especially in areas related to recommendation systems and model evaluation, would be highly valuable. I hope to have weekly meetings (0.5 to 1 hour) with my supervisor to report progress, discuss challenges, and receive constructive feedback, ensuring continuous improvement and alignment with research objectives.

7 WORKING REPORTING, RIGHTS AND OBLIGATIONS

7.1 Documentation and Publicity

All research activities, experimental results, and key insights are systematically documented in Notion to ensure transparency and easy retrieval of past work. The source code, data preprocessing scripts, and model configurations are managed using GitHub, making the research process reproducible. Weekly progress is tracked through Scrum-based updates, ensuring that milestones are met efficiently.

Once completed, the thesis will be publicly available through the university's thesis repository, making it accessible to future researchers and professionals in the field. If the findings contribute significantly to the field of recommender systems, they may be submitted to a professional journal or presented at a relevant conference.

7.2 Planned Publication

I do not plan to publicly release my thesis beyond what is required by the university. Since my research primarily involves tuning and extending the ENSFM paper rather than introducing a fundamentally new methodology, I do not see a strong need for independent publication. However, if the university requires it to be made publicly available in its repository or another designated platform, I will comply accordingly. Additionally, if there are any recommendations from my supervisor or others regarding potential publication opportunities, I am open to listening and considering them.

7.3 Copyright and Responsibilities

The copyright of this thesis will be shared among myself, my supervisor, and the university, in accordance with standard academic policies. I fully respect the intellectual property of the original ENSFM authors, and I will ensure that all references to their work are clearly cited throughout the thesis. Proper attribution will be given to the foundational research that my work builds upon, maintaining academic integrity and acknowledging the contributions of the original researchers.